RISHABH RAJ

Software Engineering • AI & Multi-LLM Systems • FinTech & Banking Technology

Puducherry, India | +91 92792 28578 | rajrishabh23959@gmail.com | rishabhraj.is-a.dev | github.com/rajrishabh23959-oss | linkedin.com/in/rishabh-raj

EDUCATION

Pondicherry University (Central University) — B.Tech + MBA (Computer Science & Business Systems / Banking Tech) **2025 – 2030 | CGPA: 9.25 / 10.0**

- **Academic Standing:** Sem 1 SGPA: **9.23/10** • Sem 2 SGPA: **9.27/10** • Subject Distinctions: **Linear Algebra (99/100)**, **Engineering Math (95/100)**, **Electronics (90/100)**, **Accounting (87/100)**.
- **Coursework & Prior Academics:** Data Structures & Algorithms (DSA), OOP, DBMS & SQL, Operating Systems, Core Banking Systems • **Class XII (BSEB):** 84.6% (**100/100 in Mathematics**) • **Class X (CBSE):** 84.8%.

TECHNICAL SKILLS

Languages & Core CS	**Python (Advanced)**, **C**, **C++**, **TypeScript**, **JavaScript (ES6+)**, **SQL**, HTML5, CSS3 • Data Structures & Algorithms, OOP, DBMS, System Design
Backend & Frontend	**FastAPI**, **Node.js**, **Express.js**, **React 18**, **Next.js 15**, **React Native**, Vite, Tailwind CSS, Pydantic v2, SQLAlchemy, Alembic, Capacitor, PyGame
AI, LLM & Security	**Multi-LLM Pipelines** (Groq Llama-3.3-70B, Mistral OCR + LLM, Gemini API), **RAG**, Prompt Engineering, Deterministic Rule Engines (Verhoeff, Mod-36, SHA-256), Behavioral Biometrics
Databases & DevOps	**PostgreSQL 15**, **MongoDB**, **Redis 7 & RQ**, Neon Serverless DB, **Docker & Docker Compose**, Nginx, **GitHub Actions (CI/CD)**, Vitest, Zod, Render, Vercel, Git

TECHNICAL PROJECTS & SYSTEM ARCHITECTURE

DualTrust AI — Dual-Pipeline AI & Deterministic Loan Verification | FastAPI, React 18, TS, PostgreSQL, Redis, Docker **Team Lead** • Architecture Docs

- **Multi-Model Consensus Architecture:** Architected a 5-container loan document verification system (**IIT Guwahati DISRUPT'27** & **IIIT Kottayam Techashy 2.0**) processing PAN, Aadhaar, salary slips, bank statements, ITR, and GST filings in parallel across two independent AI pipelines (**Groq Llama-3.3-70B** and **Mistral OCR + LLM**) mapped to a unified JSON schema.
- **Deterministic Fraud & Checksum Engine:** Engineered 10 mathematical validation rules (Aadhaar Verhoeff checksum, GSTIN Mod-36, PAN/IFSC format, salary arithmetic, duplicate SHA-256 hash, velocity checks) paired with a weighted consensus engine (40% model agreement, 30% cross-document income/identity reconciliation, 20% rule integrity, 10% OCR confidence).
- **Explainable Triage & Cryptographic Audit Log:** Automated risk-confidence triage (0–100 routing to PASS ≥90, REVIEW 70–89, HIGH RISK <70) with a React/Recharts underwriter dashboard and tamper-evident **SHA-256 hash-chained audit logs** compliant with RBI Digital Lending Guidelines (2022), cutting projected verification cost by **60–75%**.

RenoPay — Full-Stack UPI Payments, Biometric Security & Ledger Platform | FastAPI, React, PostgreSQL, Docker, Nginx GitHub • Live Demo

- **End-to-End Payment & Double-Entry Ledger:** Engineered a full-stack UPI payments ecosystem supporting real-time peer-to-peer transfers, UPI Lite micro-transactions, shared vaults, savings goals, voice-assisted entry, and an AI-powered **Digital Khatabook** module built on strict double-entry journal and ledger accounting principles.
- **Project SentinAI & Cloud CI/CD:** Embedded continuous **keystroke-dynamics behavioral biometrics** (analyzing typing rhythm, timing, and pressure to prevent account takeover beyond static passwords) and deployed containerized services via **Docker + Nginx** on Render with automated **GitHub Actions CI** security linting and test suites.

GrillRoom — AI Investor Pitch Simulation & Deck Evaluation Platform | Next.js 15, TypeScript, Tailwind, Neon DB, Zod, Vitest Live App • GitHub

- **AI Investor Q&A & PDF Deck Parsing:** Built an interactive pitch-practice platform for **PromptWars × The Prompt Arena** (ranked **7th University-Wide with 96.54 score**), enabling founders to parse PDF pitch decks (`unpdf`) and undergo real-time, multi-turn Q&A grilling against specialized AI venture-capital investor personas.
- **Schema-Validated Rubrics & Test Suites:** Designed deterministic **Zod** schema validation for structured LLM evaluation rubrics backed by **Neon Serverless PostgreSQL**, resolving offline runner build bottlenecks and establishing automated unit testing with **Vitest**, JSDOM, and GitHub Actions CI on Vercel.

MargDarshan & MAHAN — Scholarship Portal & Consulting Marketplace | Next.js, React, TS, Node.js, PostgreSQL, Gemini AI Portfolio • GitHub

- **MargDarshan (MOTA Scholarship Portal):** Built a unified application portal consolidating 5 Ministry of Tribal Affairs (MOTA) schemes with an automated certificate verification pipeline, reusable document wallet, and multilingual voice-enabled AI companion (**SAATHI**) to streamline student onboarding.
- **MAHAN (Micro-Consulting Platform):** Solo-engineered a full-stack expert session marketplace integrating **Razorpay** payment flows, **Gemini AI** matching, hybrid PostgreSQL/MongoDB persistence, and **Twilio** notifications.

EklavyaX, AgriEdge & Treasure Hunter — EdTech, AgriTech & 2D Engine | Python, FastAPI, React Native, PyGame Portfolio • GitHub

- **EklavyaX (Smart India Hackathon 2026) & AgriEdge (Schneider Electric Yuva Yodha 2026):** Architected a multi-portal gamified STEM learning platform with a custom RAG tutor (`Synapse.ai`), streak retention mechanics, and peer challenges; designed the **AgriEdge** clean-energy agricultural sustainability solution for Schneider Electric's national hackathon.
- **Treasure Hunter (2D Physics Engine):** Engineered a 2D game engine from scratch in Python & PyGame featuring decoupled physics and rendering loops, custom AABB collision detection, and modular OOP state-machine architecture.

HONORS, HACKATHONS & COMPETITIVE ACHIEVEMENTS

- **1st Rank — Travel Tech X Ideathon (Cash Prize Winner, 2026):** Clinched 1st place and cash prize in the premier travel technology innovation challenge on World Tourism Day 2026 at Pondicherry University.
- **1st Rank — Tourism AI: The Case (Cash Prize Winner, 2026):** Won 1st place and cash prize for architecting scalable AI-driven tourism solutions at Pondicherry University.
- **7th Rank University-Wide (Score: 96.54) — PromptWars × The Prompt Arena (2026):** Selected among the **Top 10 Jury Pitch Finalists** in the Prompt Engineering competition organized by the **CSE Department, Pondicherry University**, in association with **Hack2Skill** & **Google for Developers**.
- **2× JEE Advanced Qualifier (2024, 2025) & NDA Written Exam Qualifier:** Qualified twice in India's premier national engineering entrance exam (top ~1.5% nationwide, with **100/100 in Class XII Mathematics**) and cleared the UPSC National Defence Academy (NDA) written examination.
- **Certifications & National Forums: CS107: C++ Programming Certification** (82.50%, Saylor Academy, Aug 2026) • Nominated University Representative for **Smart India Hackathon (SIH 2025 & 2026)** • **Google Cloud Learn with Communities** (2026) • Pragyan NIT Trichy Technical Fest.